

FMCW/TWTT Mathematical Handbook

Equations, assumptions, units, limiting cases, and failure modes for the MATLAB model

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Curated research synthesis and engineering reference

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1 Notation and clock model

Use SI units internally. Let c denote propagation speed (use the exact speed of light in vacuum for an ideal free-space simulation). Let station A and station B have local clock readings

$$T_A(t) = (1 + \epsilon_A)t + \theta_A + \nu_A(t), \quad T_B(t) = (1 + \epsilon_B)t + \theta_B + \nu_B(t),$$

where t is an ideal reference time, θ_i is epoch offset, ϵ_i is fractional rate error, and $\nu_i(t)$ is stochastic timing jitter/wander. Relative quantities are

$$\theta = \theta_B - \theta_A, \quad \epsilon = \epsilon_B - \epsilon_A.$$

For the first ideal simulation set $\epsilon = 0$ and $\nu_i = 0$, but retain θ as the unknown to recover.

Failure mode: calling a shared oscillator “time synchronization” is incorrect. Equal rates ($\epsilon = 0$) do not imply equal epochs ($\theta = 0$), and neither condition guarantees equal RF phase.

2 Linear FMCW chirp

A convenient complex analytic chirp is

$$s(t) = A \exp \left\{ j \left[\phi_0 + 2\pi f_0 t + \pi S t^2 \right] \right\}, \quad 0 \leq t < T_c,$$

with instantaneous frequency

$$f(t) = \frac{1}{2\pi} \frac{d\phi}{dt} = f_0 + St.$$

The slope is

$$S = \frac{B}{T_c} \quad [\text{Hz/s}],$$

where B is sweep bandwidth and T_c is ramp duration.

Units check: B is Hz, T_c is s, so S is Hz/s. Multiplying S by a delay in seconds yields a beat frequency in Hz.

3 Delay and range conventions

For an active one-way link of separation d ,

$$\tau = \frac{d}{c}.$$

For a monostatic reflection at range R ,

$$\tau = \frac{2R}{c}.$$

Therefore a timing error of 10 ps corresponds to a one-way path length of roughly 3 mm; if the same 10-ps quantity were a monostatic round-trip delay it would correspond to roughly 1.5 mm of range.

4 Dechirping derivation

Let the received one-way signal be an attenuated delayed copy

$$r(t) = \alpha s(t - \tau).$$

Using the convention $z(t) = s(t)r^*(t)$,

$$\phi_z(t) = \phi(t) - \phi(t - \tau) \quad (1)$$

$$= 2\pi f_0 \tau + 2\pi S \tau t - \pi S \tau^2. \quad (2)$$

Thus the beat frequency is

$$f_b = S\tau.$$

Changing the mixer/conjugation order changes the sign. The physical estimator must use one convention consistently.

For monostatic range,

$$R = \frac{cf_b}{2S}.$$

For an ideal active one-way link,

$$d = \frac{cf_b}{S}.$$

5 Beat phase

At a chosen time origin the beat has phase

$$\phi_b(t) = 2\pi S \tau t + \phi_{b,0},$$

where

$$\phi_{b,0} = 2\pi f_0 \tau - \pi S \tau^2 + \Delta\phi_0$$

for the selected convention. When τ is small enough that the quadratic term is negligible, the phase sensitivity is approximately

$$\frac{d\phi_b}{d\tau} \approx 2\pi f_0.$$

This is extraordinarily sensitive at 77 GHz, but phase is ambiguous modulo 2π . Beat frequency provides coarse/unambiguous information; phase provides fine local correction.

6 Ideal reciprocal two-way time transfer

Assume reciprocal physical propagation delay τ and relative clock epoch offset θ . Define effective measured delays under one sign convention:

$$\delta_{AB} = \tau + \theta, \quad \delta_{BA} = \tau - \theta.$$

Then

$$\tau = \frac{\delta_{AB} + \delta_{BA}}{2}, \quad \theta = \frac{\delta_{AB} - \delta_{BA}}{2}.$$

If both directions use the same known slope S and all other offsets are zero,

$$f_{AB} = S(\tau + \theta), \quad f_{BA} = S(\tau - \theta),$$

so

$$\boxed{\hat{\tau} = \frac{f_{AB} + f_{BA}}{2S}}, \quad \boxed{\hat{\theta} = \frac{f_{AB} - f_{BA}}{2S}}.$$

Assumptions: same signed slope, reciprocal propagation, no carrier-frequency offset, no clock skew over the observation, no unequal hardware group delay, and an estimator that preserves beat sign.

7 Two-way hardware-delay asymmetry

Let the effective fixed transmit/receive delays be $g_A^{TX}, g_A^{RX}, g_B^{TX}, g_B^{RX}$. Then a simple timestamp-like model is

$$\begin{aligned} \delta_{AB} &= \tau + \theta + g_A^{TX} + g_B^{RX}, \\ \delta_{BA} &= \tau - \theta + g_B^{TX} + g_A^{RX}. \end{aligned}$$

The two-way estimates become

$$\begin{aligned} \hat{\tau} &= \tau + \frac{g_A^{TX} + g_A^{RX} + g_B^{TX} + g_B^{RX}}{2}, \\ \hat{\theta} &= \theta + \frac{(g_A^{TX} - g_A^{RX}) - (g_B^{TX} - g_B^{RX})}{2}. \end{aligned}$$

Thus reciprocal over-the-air propagation does not magically calibrate internal asymmetry. Absolute time-transfer accuracy requires calibration; repeatability/precision can be much better than uncalibrated accuracy.

8 Carrier-frequency offset

Suppose the received remote chirp has carrier offset Δf relative to the local chirp but equal slope. Then the dechirped instantaneous beat becomes approximately

$$f_b = S\tau + \Delta f$$

(up to sign convention). A single up-chirp cannot distinguish delay from CFO. Using both up and down slopes,

$$f_+ = \Delta f + S\tau, \quad f_- = \Delta f - S\tau,$$

so

$$\Delta f = \frac{f_+ + f_-}{2}, \quad \tau = \frac{f_+ - f_-}{2S}.$$

This is the core reason the early cooperative synchronization literature uses multiple slope directions.

9 Slope mismatch

Let the remote and local slopes differ by ΔS . The frequency difference is no longer constant:

$$f_b(t) \approx \Delta f + S\tau + \Delta S t + \text{small cross terms}.$$

The beat is itself chirped. A pure-tone estimator becomes biased or broadened. In simulation, slope mismatch should be represented at waveform generation, not inserted as arbitrary additive beat noise.

10 Clock skew

If relative clock offset evolves as

$$\theta(t) = \theta_0 + \epsilon t,$$

then an FMCW timing beat caused by θ drifts in time. To first order, substituting $\theta(t)$ into $S(\tau \pm \theta)$ produces a linear beat-frequency drift proportional to $S\epsilon$. Sample-clock frequency error also rescales the digital time axis, causing an additional estimator bias. Keep waveform-clock skew and ADC/sample-clock skew as separate parameters if the hardware architecture permits them to differ.

11 Doppler

For a narrowband approximation with radial velocity v , Doppler frequency is

$$f_D \approx \frac{2v}{\lambda}$$

for a monostatic reflection and depends on bistatic geometry for a distributed link. In ordinary sawtooth FMCW,

$$f_b \approx S\tau + f_D.$$

Up/down ramps can help separate range and Doppler because the range contribution changes sign with slope while Doppler does not under the conventional formulation.

For the first TWTT simulation, set $v = 0$. Add Doppler only after the clock-offset algebra is validated.

12 Sampling and the FFT

With sample rate F_s , N samples, and observation duration $T_{obs} = N/F_s$, the DFT bin spacing is

$$\Delta f_{DFT} = \frac{F_s}{N} = \frac{1}{T_{obs}}.$$

Zero padding makes a denser display/interpolation grid but does not create new information. A sinusoid can nevertheless be estimated to a small fraction of Δf_{DFT} when SNR and model fidelity are adequate.

For $F_s = 10$ MS/s and $N = 256$,

$$T_{obs} = 25.6 \text{ } \mu\text{s}, \quad \Delta f_{DFT} = 39.0625 \text{ kHz}.$$

At $S = 29.982$ MHz/ μs ,

$$10 \text{ ps} \rightarrow 299.82 \text{ Hz}, \quad 100 \text{ ps} \rightarrow 2.9982 \text{ kHz}, \quad 1 \text{ ns} \rightarrow 29.982 \text{ kHz}.$$

So 10 ps is far below one raw FFT bin in the current short observation. That is an estimator-design problem, not an algebra error.

13 Phase-slope frequency estimator

For a high-SNR complex tone

$$z[n] = Ae^{j(2\pi f_b n/F_s + \phi_0)} + w[n],$$

unwrapping phase and fitting a straight line gives

$$\angle z[n] \approx \frac{2\pi f_b}{F_s} n + \phi_0.$$

A least-squares slope $\hat{m}$ yields

$$\hat{f}_b = \frac{F_s}{2\pi} \hat{m}.$$

This is a useful first sub-bin estimator because it is transparent and easy to unit-test. It is not optimal under all noise/distortion conditions, so later compare it with CZT/interpolated-DFT and nonlinear sinusoid fits.

14 CZT refinement

The chirp-z transform evaluates the z -transform along a user-chosen contour and can sample a narrow frequency interval much more densely than a full-band FFT. The Scherr paper uses this property to refine FMCW beat frequency efficiently and then evaluates phase. In the simulator:

1. use FFT to find a coarse peak region;
2. place a narrow CZT interval around it;
3. estimate the refined frequency;
4. evaluate complex phase at the refined frequency;
5. combine coarse range and fine phase only after ambiguity handling is defined.

15 Generic sinusoid frequency CRLB scaling

For a single complex sinusoid in white Gaussian noise, the exact Cramer-Rao expression depends on the chosen SNR and amplitude/phase nuisance-parameter convention. A commonly encountered scaling is

$$\text{var}(\hat{f}) \propto \frac{F_s^2}{\text{SNR } N(N^2 - 1)}.$$

The important project-level implications are robust even when convention-dependent constants differ:

- precision improves rapidly with coherent observation length;
- precision improves with SNR;
- mapping frequency error to timing error divides by slope, $\sigma_\tau \approx \sigma_f/|S|$;
- colored phase noise, nonlinearity, multipath, and model mismatch can dominate long before the white-noise CRLB is reached.

The simulation report should quote the exact CRLB formula only alongside the estimator/noise convention used.

16 Delay-estimation bandwidth intuition

For a known waveform in white noise, time-delay CRLBs scale inversely with SNR and the square of an effective/root-mean-square signal bandwidth. This is the deeper reason both UWB pulses and spectrally separated two-tone synchronization can achieve excellent timing precision. FMCW gives a practical way to exploit a wide frequency excursion while keeping the post-dechirp ADC burden low.

17 Thermal and receiver noise

At an input-equivalent noise temperature T_0 and bandwidth B_n , available thermal noise power is approximately

$$P_n = kT_0B_n.$$

Receiver noise figure increases the effective noise floor. In a sampled complex-baseband truth model, the simplest correct first step is to inject complex AWGN at a specified post-dechirp SNR. A hardware-faithful model should instead propagate gain/noise figure and IF filter bandwidth to the ADC.

18 ADC quantization

For an ideal N_b -bit full-scale sinusoidal ADC, the familiar quantization-limited SNR approximation is

$$\text{SNR}_q \approx 6.02N_b + 1.76 \text{ dB},$$

but real ENOB and the actual signal level should be used when available. Quantization should be modeled after analog/IF filtering and scaling, not as an arbitrary equivalent range error.

19 Phase-noise model

Write the transmitted phase as

$$\phi_{TX}(t) = \phi_{ideal}(t) + \phi_{sys}(t) + \phi_{PN}(t).$$

A useful systematic model is

$$\phi_{sys}(t) = \sum_{k=0}^K a_k t^k + A_s \sin(2\pi f_s t + \psi_s) + \dots$$

and $\phi_{PN}(t)$ should be generated from a colored PSD when precision matters. The Tschapek workflow is: characterize PSD/systematic terms, synthesize virtual distorted chirps, run Monte Carlo radar estimation, and compare precision metrics.

20 Range correlation of oscillator phase noise

In a homodyne monostatic system using the same oscillator for TX and LO, the phase-noise contribution after dechirp has the difference form

$$\phi_{PN,out}(t) = \phi_{PN}(t) - \phi_{PN}(t - \tau).$$

In frequency domain the differencing has magnitude-squared response

$$|1 - e^{-j2\pi f\tau}|^2 = 4 \sin^2(\pi f\tau).$$

Low-offset-frequency phase noise is therefore strongly suppressed for sufficiently short delay. With independent TX and LO oscillators, the noise terms are not correlated and this cancellation is largely lost. Reciprocal/full-duplex processing can restore cancellation of some common structured terms, which is central to CFDDS-style methods.

21 Chirp nonlinearity

Let instantaneous frequency be

$$f(t) = f_0 + St + e_f(t).$$

Then phase contains the integrated error

$$\phi(t) = \phi_0 + 2\pi f_0 t + \pi S t^2 + 2\pi \int_0^t e_f(u) du.$$

The beat frequency becomes time dependent according to the difference between error trajectories sampled at t and $t - \tau$. Therefore simulate nonlinearity at the chirp phase/frequency level. Useful $e_f(t)$ test cases are:

- quadratic/cubic deterministic curvature;
- sinusoidal spur modulation;
- measured residual error trace;
- random colored frequency error tied to phase-noise PSD.

22 IF filtering and group delay

An IF filter with transfer function $H(f) = |H(f)|e^{j\phi_H(f)}$ has group delay

$$\tau_g(f) = -\frac{1}{2\pi} \frac{d\phi_H(f)}{df}.$$

If τ_g differs between stations or directions, a reciprocal air path can still yield biased two-way timing. The hardware model should therefore support per-station complex IF transfer functions or at least constant + linear group-delay approximations.

23 Multipath

For L propagation paths,

$$r(t) = \sum_{\ell=1}^L \alpha_\ell s(t - \tau_\ell) e^{j\psi_\ell}.$$

After dechirp, multiple paths become multiple beat components. If they are unresolved, the observed complex tone is a vector sum whose phase/frequency estimate can be biased. This is particularly dangerous for picosecond timing because a weak nearby path can perturb phase substantially even when the coarse range peak looks clean.

24 Phase-coded FMCW

Let a unit-modulus phase code be

$$c(t) = e^{j\phi_c(t)}.$$

A coded chirp is

$$s_c(t) = s(t)c(t).$$

A delayed return contains $c(t - \tau)$ while local decoding uses some reference $c(t)$. Their misalignment creates a delay-dependent code term. The first conceptual simulation can correlate/realign the code in

discrete time. A more faithful PC-FMCW model should include the group-delay filtering or equivalent alignment described in the phase-coded FMCW literature and verify that the required baseband sampling rate remains realistic.

25 Processing gain and coherent integration

For M repeated coherent observations with identical phase relation and independent additive noise, coherent summation ideally increases signal amplitude by M and noise RMS by $\sqrt{M}$, giving an SNR gain of M (or $10 \log_{10} M$ dB). This ideal gain fails when inter-chirp phase is not stable. Therefore multi-chirp timing precision is inseparable from chirp-to-chirp phase coherence.

26 AWR2944 parameter mapping

The MATLAB parameter struct should distinguish conceptual and hardware fields:

Math parameter	Hardware/config analogue	Note
f_0	profile start frequency	RF carrier/chirp start
S	profile frequency slope	convert MHz/us to Hz/s exactly
T_c	ramp end/active time	ADC window may be shorter than total ramp
F_s	ADC sample rate	baseband sample rate, not RF carrier sampling
N	ADC samples/chirp	sets raw observation window
B_{IF}	analog/digital IF bandwidth	constrains usable beat frequency
M	chirps/frame / coherent set	phase coherence must be verified
$\phi_{TX,k}$	TX phase shifter settings	relevant to MIMO, not timing by itself
reference clock	external clock input	syntonization does not guarantee absolute time/phase
trigger timing	frame/chirp event mechanism	must be characterized in two-board setup

27 Numerical sanity checks for the current slope

Using $S = 29.982 \times 10^{12}$ Hz/s:

$$S(10 \text{ ps}) = 299.82 \text{ Hz},$$

$$S(100 \text{ ps}) = 2.9982 \text{ kHz},$$

$$S(1 \text{ ns}) = 29.982 \text{ kHz},$$

$$S(5 \text{ ns}) = 149.91 \text{ kHz}.$$

For an illustrative two-way case $\tau = 5 \text{ ns}$, $\theta = 100 \text{ ps}$,

$$f_{AB} = S(5.1 \text{ ns}) \approx 152.9082 \text{ kHz},$$

$$f_{BA} = S(4.9 \text{ ns}) \approx 146.9118 \text{ kHz}.$$

Their average divided by S gives 5 ns, and half-difference divided by S gives 100 ps.

28 Simulation invariant checklist

Every time an impairment is added, rerun the following baseline tests:

1. $\tau = 0, \theta = 0$;
2. known fractional τ not aligned to sample spacing;
3. slope scaling test $S \rightarrow 2S$;
4. sign-convention test for conjugation/mixer order;
5. two-way sum/difference test;
6. noise-free estimator reaches numerical precision;
7. all impairment switches off returns the exact ideal result;
8. each impairment on by itself matches an analytic first-order prediction.

29 What not to do

- Do not sample a 77-GHz real cosine at hundreds of gigasamples/s for the first model.
- Do not implement 10-ps delay by rounding to the nearest ADC sample.
- Do not equate FFT-bin width with estimator precision.
- Do not combine CFO, clock skew, trigger offset, and phase noise into one “clock error” scalar.
- Do not claim two-way cancellation removes fixed hardware asymmetry unless it is modeled/calibrated.
- Do not add phase coding before the uncoded two-way algebra passes unit tests.

30 Reference anchors

The equations and modeling choices in this handbook are anchored primarily to the Roehr 2007 cooperative synchronization paper, Scherr 2015 frequency/phase estimator paper, Herzel 2018 PLL precision analysis, Gottinger 2019 CFDDS-TWR paper, Tschapek 2022 phase-distortion model, Scheibhofer 2008 simulator architecture, classical two-way time-transfer references, and TI AWR2944/chirp documentation. Exact sign conventions should be re-derived in code from the chosen complex mixer definition rather than copied mechanically from any one source.

31 General two-oscillator chirp derivation

A more realistic two-node model gives each station its own carrier, slope, phase, and local time. Let

$$\begin{aligned}\phi_A(t) &= \phi_{A0} + 2\pi f_A T_A(t) + \pi S_A T_A^2(t) + \eta_A(t), \\ \phi_B(t) &= \phi_{B0} + 2\pi f_B T_B(t) + \pi S_B T_B^2(t) + \eta_B(t),\end{aligned}$$

where η_i groups stochastic and systematic phase distortions. The signal transmitted by A and received at B is evaluated at physical time $t - \tau_{AB}$ but mixed against B's current local oscillator. Ignoring amplitude for clarity, a cross-link beat phase can be represented as

$$\Phi_{AB}(t) = \phi_B(t) - \phi_A(t - \tau_{AB}) + \phi_{ch,AB}(t).$$

Differentiating gives the instantaneous beat frequency. This equation is the safest general starting point because ideal delay, CFO, clock skew, slope mismatch, phase noise, Doppler/channel phase, and group-delay corrections can all be introduced explicitly before simplifying.

31.1 First-order expansion

Around small relative errors, write $f_B = f_A + \Delta f$, $S_B = S_A + \Delta S$, and $T_B(t) \approx t + \theta + \epsilon t$. Keeping only first-order terms produces a beat containing a constant delay term $S\tau$, a constant/slow CFO term Δf , a slope-mismatch drift $\Delta S t$, and clock-skew-dependent drift. The exact signs depend on mixer convention. In code, derive the expected sign numerically from the general phase equation and freeze it in unit tests.

32 Four timestamps and canonical two-way transfer

Classical two-way time transfer is often written with four local timestamps:

- t_1 : A transmits according to clock A;
- t_2 : B receives according to clock B;
- t_3 : B transmits response according to clock B;
- t_4 : A receives according to clock A.

For reciprocal propagation and negligible motion over the exchange, the conventional offset/range combinations can be formed from $(t_2 - t_1)$ and $(t_4 - t_3)$. FMCW replaces direct timestamping of an edge with an estimate of the effective relative delay embedded in phase/frequency. The same calibration logic still applies.

33 Nonreciprocal propagation

Let forward and reverse propagation delays be

$$\tau_{AB} = \tau + \frac{\Delta\tau_{nr}}{2}, \quad \tau_{BA} = \tau - \frac{\Delta\tau_{nr}}{2}.$$

Then the ideal two-way offset solution is biased by

$$\hat{\theta} = \theta + \frac{\Delta\tau_{nr}}{2}$$

(up to sign convention). Any motion between the two directional measurements, different RF frequencies, polarization-dependent paths, or asymmetric hardware/channel group delay can therefore masquerade as clock offset.

34 Moving nodes during reciprocal exchange

If separation changes as $d(t) = d_0 + vt$, propagation delay changes by approximately $v\Delta t/c$ across an exchange interval Δt . For a 10-ps objective, even small physical motion can matter if the two directions are not simultaneous. Full-duplex double-sided methods reduce this temporal separation compared with sequential request/response schemes. A simulation should therefore include the exact transmit times of each directional chirp when motion is enabled.

35 Phase ambiguity and synthetic wavelength

A phase measurement at carrier f_c is periodic in delay with period

$$\Delta\tau_{2\pi} = \frac{1}{f_c},$$

which corresponds to a one-way distance period $c/f_c = \lambda$. Combining phases at two frequencies f_1 and f_2 produces a beat/synthetic wavelength

$$\Lambda = \frac{c}{|f_2 - f_1|}.$$

This is another way to understand why multi-frequency/two-tone methods can achieve fine phase sensitivity while extending unambiguous range. FMCW continuously sweeps many frequencies and uses the beat slope/frequency to resolve coarse delay.

36 Frequency/phase fusion

Suppose a coarse estimate τ_c from beat frequency identifies the correct phase ambiguity interval. A fine phase residual $\Delta\phi$ at effective carrier f_{eff} gives approximately

$$\Delta\tau_\phi = \frac{\Delta\phi}{2\pi f_{eff}}.$$

A fused estimate can be written

$$\hat{\tau} = \tau_c + \Delta\tau_\phi,$$

provided τ_c is accurate enough to choose the correct integer phase cycle. In practice, f_{eff} and constant phase terms should be derived from the full chirp phase equation rather than assuming exactly f_0 .

37 Phase unwrap by model fitting

For a complex beat observed at samples n , a robust phase model is

$$\phi[n] = \phi_0 + \omega n + q[n],$$

where $q[n]$ contains residual nonlinearity/noise. Rather than unwrapping point-by-point when SNR is low, one can fit the complex exponential directly:

$$\min_{A, \phi_0, \omega} \sum_n \left| z[n] - A e^{j(\omega n + \phi_0)} \right|^2.$$

This nonlinear least-squares/maximum-likelihood viewpoint is useful because it handles frequency and phase jointly and avoids some unwrap failures. It is computationally heavier than a simple phase slope or CZT.

38 Windowing and spectral leakage

If a finite observation does not contain an integer number of beat cycles, a rectangular window spreads energy through sinc sidelobes. A window $w[n]$ changes both main-lobe width and estimator variance. For plot-only FFTs, Hann-like windows reduce sidelobes; for precision estimation, the chosen estimator and its CRLB should account for the window. Never compare two estimator variances while silently changing window functions.

39 Zero padding

With zero padding factor K , the FFT display grid becomes

$$\Delta f_{grid} = \frac{F_s}{KN}.$$

The underlying observation time remains N/F_s . Thus zero padding can improve peak interpolation and visual presentation but does not by itself create the information of a KN -sample observation. This distinction should be demonstrated in one simulation figure because it is pedagogically valuable.

40 White-noise frequency CRLB with explicit convention

For the model

$$x[n] = Ae^{j(\omega n + \phi)} + w[n], \quad w[n] \sim \mathcal{CN}(0, \sigma^2),$$

and defining complex-sample SNR as $\rho = A^2/\sigma^2$, one common form for unknown amplitude/phase gives a frequency-in-radians/sample variance proportional to

$$\text{var}(\hat{\omega}) \geq \frac{6}{\rho N(N^2 - 1)}.$$

Converted to hertz,

$$\text{var}(\hat{f}) \geq \left(\frac{F_s}{2\pi}\right)^2 \frac{6}{\rho N(N^2 - 1)}.$$

Different noise/SNR definitions can introduce factors of two, so code and report must state the convention. The corresponding ideal timing standard deviation is

$$\sigma_\tau \geq \frac{\sigma_f}{|S|}.$$

Use this only as a white-noise benchmark; it is not a bound for a misspecified colored-phase-noise model.

41 Fisher information for delay

For a known deterministic complex waveform $s(t - \tau)$ in AWGN, delay Fisher information is proportional to the energy of its time derivative,

$$\mathcal{I}_\tau \propto \frac{1}{N_0} \int \left| \frac{ds(t)}{dt} \right|^2 dt.$$

In frequency domain this weights signal energy by frequency squared. After removing an arbitrary carrier-phase nuisance, the relevant quantity is related to rms bandwidth. This formalizes the intuitive statement that widely separated frequencies carry strong delay information.

42 Allan deviation and clock stability

Clock frequency stability is often described by fractional-frequency fluctuations $y(t)$ and Allan deviation $\sigma_y(\tau_a)$ over averaging time τ_a . A constant fractional frequency error ϵ causes time offset to grow approximately linearly, $\theta(t) = \theta_0 + \epsilon t$. Stochastic oscillator noise produces different power-law growth depending on noise type. For long-duration synchronization simulations, specifying only a one-shot CFO is insufficient; an Allan-deviation/power-law clock model may be more appropriate.

For the first chirp-scale model, however, a constant CFO + white/colored phase-noise process is enough. Do not introduce Allan-noise complexity before the estimator is validated.

43 Phase-noise PSD synthesis algorithm

Given a target single-sideband phase-noise mask $L(f)$ in dBc/Hz, a practical numerical workflow is:

1. convert the chosen convention to a phase PSD $S_\phi(f)$ in rad^2/Hz ;
2. create a symmetric frequency grid compatible with the simulation record length;
3. draw complex Gaussian frequency-domain samples with variance proportional to $S_\phi(f)\Delta f$ and Hermitian symmetry for a real phase process;
4. inverse FFT to obtain $\phi_{PN}(t)$;
5. verify the generated PSD by ensemble averaging;
6. add $\phi_{PN}(t)$ to phase, not to amplitude.

Be explicit about one-sided versus two-sided PSD and the relation between SSB $L(f)$ and phase PSD. This is an easy place to make a 3-dB or factor-of-two mistake.

44 Correlated versus independent phase noise

For two oscillators with phase processes ϕ_A and ϕ_B , allow a correlation coefficient or shared/reference component:

$$\phi_A = \phi_c + \phi_{A,u}, \quad \phi_B = \phi_c + \phi_{B,u}.$$

A shared reference can reduce low-frequency relative phase/frequency noise without making the high-frequency PLL/VCO noise identical. Modeling separate common and uncorrelated components is more realistic than either “same phase noise” or “completely independent” when the boards share a low-frequency reference.

45 PLL settling and ramp transients

A fast frequency ramp through a PLL is not necessarily an exact parabola in phase. Loop bandwidth and charge-pump/synthesizer dynamics can cause settling/overshoot near segment boundaries. If the ADC valid window begins after a settling interval, much of this transient may be excluded. Therefore the AWR-faithful simulator should model the actual ADC start time relative to ramp start before adding a detailed PLL transient model.

46 Near-DC beat considerations

A timing-induced beat of only hundreds of hertz can fall close to DC where analog high-pass filters, flicker noise, LO leakage, DC offsets, and digital clutter-removal algorithms are strongest. A deliberate known frequency offset f_{off} can move the beat away from DC:

$$f_b = f_{off} + S\tau.$$

After estimating f_b , subtract f_{off} in processing. The offset itself must be stable/calibrated and should not push the beat outside the IF bandwidth.

47 Coded-waveform correlation

For discrete code sequence $c_i[k]$ assigned to station i , code identification can use a normalized correlation

$$R_i[m] = \frac{\sum_k r[k]c_i^*[k-m]}{\sqrt{\sum_k |r[k]|^2 \sum_k |c_i[k]|^2}}.$$

Useful code families have low cross-correlation and controlled sidelobes. In phase-coded FMCW the code is embedded in chirp phase, so delay and dechirp alter the code alignment. Correlation performance must therefore be evaluated after the actual dechirp/alignment chain, not just on isolated ideal code vectors.

48 Monte Carlo statistics

For M trials with estimates $\hat{\theta}_m$ and truth θ ,

$$\begin{aligned} \text{bias} &= \frac{1}{M} \sum_m (\hat{\theta}_m - \theta), \\ \sigma &= \sqrt{\frac{1}{M-1} \sum_m (\hat{\theta}_m - \bar{\hat{\theta}})^2}, \\ \text{RMSE} &= \sqrt{\frac{1}{M} \sum_m (\hat{\theta}_m - \theta)^2}. \end{aligned}$$

Report all three when systematic errors are present. A low standard deviation can coexist with a large bias.

49 Confidence interval on estimated standard deviation

For Gaussian independent errors, uncertainty in a measured variance decreases slowly with trial count. A few dozen Monte Carlo runs are inadequate for credible ps-level tail/variance claims. Use at least thousands of trials for initial curves and increase for final figures, or compute confidence intervals based on chi-squared statistics.

50 Error propagation through two-way algebra

If directional delay estimates have independent variances σ_{AB}^2 and σ_{BA}^2 , then ideal sum/difference estimates have

$$\text{var}(\hat{\tau}) = \text{var}(\hat{\theta}) = \frac{\sigma_{AB}^2 + \sigma_{BA}^2}{4}.$$

If errors are correlated, covariance contributes with opposite signs:

$$\text{var}(\hat{\tau}) = \frac{\sigma_{AB}^2 + \sigma_{BA}^2 + 2 \text{cov}}{4},$$

$$\text{var}(\hat{\theta}) = \frac{\sigma_{AB}^2 + \sigma_{BA}^2 - 2 \text{cov}}{4}.$$

This is why structured common errors can cancel in the clock-offset combination while reinforcing in the range combination, or vice versa.

51 Precision versus accuracy budget template

Represent the final result as

$$\hat{\theta} = \theta + b_{cal} + b_{nr} + b_{model} + n_{est},$$

where b_{cal} is residual calibration bias, b_{nr} is nonreciprocity/asymmetry bias, b_{model} is estimator/model mismatch, and n_{est} is stochastic estimator noise. A paper claiming 10-ps *accuracy* must bound the bias terms, not merely show $\sigma(n_{est}) < 10$ ps.

52 Useful order-of-magnitude conversions

Quantity	Equivalent	Interpretation
1 ps one-way time	0.2998 mm path	tiny mechanical/electrical changes matter
10 ps one-way time	2.998 mm path	target discussed in meeting notes
100 ps one-way time	29.98 mm path	easier first hardware milestone
1 degree at 77 GHz	about 36 fs phase-time	carrier phase is extraordinarily sensitive but ambiguous
1 ppm clock-rate error over 1 ms	1 ns time drift	even ppm-class skew matters over long exchange windows
1 kHz beat error at 29.982 MHz/us	33.35 ps timing error	convenient estimator conversion